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Quantum-resistant intrusion detection in federated IoT networks: A lattice-based approach

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ABSTRACT

The proliferation of Internet of Things (IoT) devices has created unprecedented security challenges in network infrastructure. Simultaneously, the advent of quantum computing threatens the cryptographic foundations of existing Intrusion Detection Systems (IDS). Traditional centralized IDS architectures are inadequate for distributed IoT environments due to privacy concerns, communication overhead, and single points of failure. This paper presents PQ-FedIDS, a novel quantum-resistant federated intrusion detection framework that combines lattice-based cryptography with Byzantine-resilient Federated Learning (FL) for secure network anomaly detection in heterogeneous IoT networks. Our approach leverages CRYSTALS-Dilithium digital signatures for authentication and ML-KEM (Module-Lattice-Based Key-Encapsulation Mechanism) encryption for confidential model aggregation. This ensures post-quantum security guarantees with 2^{128} classical security equivalence. We implement a hierarchical privacy-preserving FL protocol with reputation-weighted multi-Krum aggregation. This enables IoT edge devices to collaboratively train intrusion detection models without exposing raw network traffic data while defending against model poisoning attacks. Experimental evaluation on three standard IDS benchmark datasets (CICIDS2017, NSL-KDD, UNSW-NB15) demonstrates that PQ-FedIDS achieves 96.8% detection accuracy with minimal cryptographic overhead (1.1% communication increase) and practical resource consumption. Formal security analysis proves post-quantum authentication unforgeability and IND-CCA2 confidentiality, while Byzantine resilience maintains 94.7% accuracy under 30% compromised devices.

1. Introduction

The exponential growth of Internet of Things (IoT) deployments has fundamentally transformed network security landscapes. According to recent industry forecasts [1], the total number of connected IoT devices is expected to grow dramatically over the coming decade, potentially exceeding 40 billion by 2030 as IoT adoption expands across manufacturing, healthcare, transportation, and energy sectors. This heterogeneity introduces unique challenges, as IoT networks comprise devices with vastly different computational capabilities, from resource-constrained sensors with limited memory to powerful gateways, making uniform security policy enforcement impractical [2].

This massive proliferation introduces severe security vulnerabilities, with network-based attacks on IoT infrastructure increasing by 300%

annually [3]. High-profile incidents, including the Mirai botnet attack that compromised more than 600,000 IoT devices [4] and vulnerabilities in medical IoT systems, demonstrate the catastrophic consequences of inadequate security measures. Traditional intrusion detection systems (IDS) face three critical limitations in IoT environments: centralized architectures create single points of failure and violate privacy requirements, privacy regulations such as GDPR prohibit raw traffic data transmission to centralized servers, and computational constraints at edge devices limit the deployment of sophisticated detection mechanisms.

At the same time, the advancement of quantum computing is an existential threat to existing cryptographic protocols. The algorithm developed by Shor enables quantum computers to crack RSA and elliptic curve encryption in a polynomial amount of time, compromising the authentication and confidentiality mechanisms underlying existing

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IDS [5]. While large-scale quantum computers remain years away, the “harvest now, decrypt later” attack paradigm poses immediate risks; adversaries can record encrypted traffic today and decrypt it once quantum computers become available [6]. To address this quantum threat, to date, the National Institute of Standards and Technology (NIST) has standardized post-quantum cryptographic algorithms such as CRYSTALS-Dilithium for digital signatures and ML-KEM for key encapsulation [7]. Nevertheless, there is an open challenge of integrating these post-quantum primitives into distributed IDS architectures, particularly for resource-constrained IoT environments where signature sizes and computational overhead substantially exceed classical alternatives.

Federated Learning (FL) is a recent, promising paradigm for privacy-preserving collaborative machine learning, which allows distributed model training without sensitive data being centralized [8]. In federated IDS, edge devices train local intrusion detection models on private network traffic and transmit only model updates to aggregators, addressing privacy concerns while enabling learning from diverse attack patterns. Recent papers have considered federated IDS for IoT networks [9,10], but current solutions do not offer quantum-resistant security guarantees and fail to mitigate authentication vulnerabilities in model aggregation protocols. Furthermore, existing approaches are susceptible to Byzantine attacks where compromised devices submit malicious model updates designed to degrade detection accuracy [11]. Model poisoning attacks, a primary Byzantine threat in federated learning, involve adversaries submitting crafted gradient updates (e.g., gradients computed on mislabeled data or scaled by large factors) to bias the globally aggregated model toward increased false negatives, enabling attacks to evade detection while maintaining plausible overall accuracy [12].

Existing federated IDS frameworks address either privacy preservation through secure aggregation [13] or Byzantine resilience through robust aggregation rules [11], but fail to provide quantum-resistant security guarantees. Prior secure aggregation protocols rely on classical Diffie-Hellman key exchange, which is vulnerable to Shor’s algorithm, making them unsuitable for long-term security against quantum adversaries. Recent Byzantine-resilient approaches, such as Krum [11], provide robustness but lack cryptographic authentication, enabling impersonation attacks where adversaries submit malicious updates under honest device identities. This work uniquely integrates NIST-standardized post-quantum primitives (CRYSTALS-Dilithium for authentication, ML-KEM for confidentiality) with adaptive reputation-weighted Byzantine-resilient aggregation, providing the first federated IDS framework with provable quantum resistance, authenticated model updates, and adaptive adversary detection. The intersection of Post-Quantum Cryptography (PQC) and FL in network security represents a critical research gap that we aim to address in the present paper.

This paper introduces PQ-FedIDS, a quantum-resistant federated intrusion detection framework designed for heterogeneous IoT networks. Our key contributions are:

- We propose a hierarchical architecture integrating lattice-based cryptography with federated learning for quantum-resistant network intrusion detection in IoT environments, featuring fog-assisted aggregation that reduces communication overhead.
- We design a secure model aggregation protocol using CRYSTALS-Dilithium signatures and ML-KEM encryption that prevents model poisoning attacks while maintaining post-quantum security guarantees with minimal overhead.
- We develop an adaptive Byzantine detection mechanism based on reputation-weighted multi-Krum aggregation that provides stronger resilience than static aggregation rules, maintaining accuracy even under a 30% Byzantine ratio.
- We provide mathematical formalization of security guarantees against quantum adversaries and Byzantine participants with formal proofs of resistance bounds, demonstrating authentication unforgeability and IND-CCA2 confidentiality.

- We conduct a comprehensive experimental evaluation demonstrating 96.8% detection accuracy with minimal cryptographic overhead (1.1% communication increase) and practical resource consumption suitable for battery-powered IoT devices.

The remainder of this paper is organized as follows. Section 2 reviews related work on federated IDS, post-quantum cryptography, and secure aggregation protocols. Section 3 presents the threat model and system architecture. Section 4 details the cryptographic protocols and FL algorithms of the proposed PQ-FedIDS framework. Section 5 provides a security analysis with formal proofs. Section 6 presents experimental evaluation results. Section 7 discusses limitations and future work, and Section 8 concludes the paper.

2. Related work

Prior research addresses federated IDS, PQC in IoT, and secure aggregation separately. We review each area, identify gaps that PQ-FedIDS addresses, and present a comparative analysis highlighting our unique contributions.

2.1. Federated learning for intrusion detection

Some recent studies have explored FL approaches for distributed IDS in the IoT-enabled systems. Liu et al. [19] present a detailed survey of FL-based IDS designs, classifying existing schemes by aggregation strategy, data partitioning, and model architecture, and describing the challenges posed by communication overhead and poisoning attacks. Hendaoui et al. [14] proposed FLADEN, a federated anomaly detection framework for IoT networks that trains lightweight local models at edge gateways and aggregates their parameters on a central server, achieving competitive detection accuracy without sharing raw data. Nguyen et al. [15] developed DIoT, a federated self-learning anomaly detection system for IoT networks that detects network and physical attacks, achieving 95% F1-score but lacking post-quantum cryptographic protection. An optimal FL-based IDS for IoT was proposed by Karunamurthy et al. [16], which combines feature selection with federated aggregation, reporting over 92% detection accuracy. Rahman et al. [17] presented a blockchain-managed FL framework for secure IoT health applications with provenance tracking, utilizing blockchain for data integrity but introducing computational overhead without addressing quantum threats. None of these approaches, however, explicitly address post-quantum security threats or provide quantum-resistant authentication within the FL-based IDS pipeline.

2.2. Post-quantum cryptography in IoT

Several studies have explored PQC for securing resource-constrained IoT devices. Kannwischer et al. [20] introduce pqm4, a comprehensive framework that implements and benchmarks multiple NIST PQC key-encapsulation and signature schemes on ARM Cortex-M4 microcontrollers, providing detailed cycle counts and stack usage measurements for signature schemes and KEMs, including implementations relevant to IoT deployments. The framework demonstrates that CRYSTALS-Dilithium signing requires 1.8M cycles (approximately 2.1 ms at 850 MHz) and ML-KEM encapsulation requires 0.9M cycles, establishing feasibility for IoT gateways but presenting challenges for ultra-low-power sensor nodes.

Schöffel et al. [21] evaluate post-quantum primitives on Cortex-M4 edge devices, demonstrating that quantum-resistant security can be provided by carefully chosen PQC algorithms at energy consumption comparable to that of classical public-key algorithms on devices when deployed with sufficient computational power. Their energy analysis shows that Dilithium signing requires approximately 3.2 mJ on battery-powered sensors at 0.5W, and ECDSA signing requires 0.8 mJ, which is 4× higher than that of gateway devices, but must be carefully considered for battery-operated sensors.

Table 1
Comparison with State-of-the-Art federated IDS approaches.

Approach	Post-Quantum Security	Byzantine Resilient	Privacy Preserving	IoT Optimized	Multi-Dataset Evaluation	Crypto Auth
FLADEN [14]	✗	✗	✓	✓	✗	✗
DIoT [15]	✗	✗	✓	✓	✗	✗
Karunamurthy [16]	✗	✗	✓	✓	✗	✗
Rahman [17]	✗	✓	✓	✗	✗	✓
Bonawitz [18]	✗	✗	✓	✗	–	✗
Krum [11]	✗	✓	✗	✗	–	✗
PQ-FedIDS (Ours)	✓	✓	✓	✓	✓	✓

NewHope, a ring-LWE-based post-quantum key-exchange protocol, was proposed by Alkim et al. [22] and demonstrated a practical key-generation time of 1.3 ms on standard processors, establishing the feasibility of lattice cryptography on resource-constrained devices. NewHope provides post-quantum security equivalent to AES-128, with public and ciphertext sizes of 1824 and 2048 bytes, respectively. However, these sizes exceed typical IoT packet payloads, which often require fragmentation.

Liu et al. [23] survey PQC in IoT ecosystems, analyze computational and communication overheads, and discuss post-quantum TLS designs that fit within constrained hardware. One of the main bottlenecks they identify is signature size. Dilithium signatures (3293 bytes) exceed typical IoT packet sizes (127 bytes for 802.15.4 packets). As a result, they must be fragmented, which increases transmission energy costs by 40–60% and introduces additional latency for reassembly.

However, these works focus on cryptographic primitives and benchmarking without addressing federated machine learning security, Byzantine resilience in distributed IDS, or the unique challenges of securing model aggregation protocols against quantum adversaries in resource-constrained environments.

2.3. Secure aggregation protocols

Bonawitz et al. [18] introduced secure aggregation for FL using multi-party computation, providing privacy against honest-but-curious servers. Their protocol enables a server to compute the sum of user-submitted vectors without learning individual contributions. However, the protocol requires $O(N^2)$ pairwise communications during the setup phase, incurs substantial computational overhead for cryptographic operations, and relies on classical Diffie-Hellman.

Bell et al. [24] proposed secure aggregation with dropout resilience, handling participant failures in federated settings through threshold secret sharing schemes that maintain privacy guarantees even when subsets of participants disconnect. The approach tolerates up to t dropped participants while preserving privacy but increases communication overhead by factor of $t + 1$ and computational cost proportional to the number of surviving participants.

So et al. [25] developed Turbo-Aggregate for communication-efficient secure aggregation, reducing bandwidth requirement from $O(N)$ to $O(\sqrt{N})$ using structured linear codes and rotation-based techniques. While achieving significant communication savings, the protocol remains vulnerable to quantum attacks due to its reliance on classical cryptographic primitives.

Blanchard et al. [11] propose the Krum aggregation rule that selects the update closest to the majority based on Euclidean distance scores computed over all pairs of submitted updates. Krum provides provable Byzantine resilience when $f < (n - 2)/2$ malicious participants are present in the aggregation cluster, but suffers from two limitations: it selects a single update per round, discarding information from all other honest participants, leading to slower convergence and higher variance in detection accuracy. It treats all participants equally, regardless of historical behavior, enabling intermittent Byzantine attackers to evade detection by occasionally submitting honest-looking updates. Our

reputation-weighted multi-Krum aggregation addresses both limitations by selecting $m = n - f$ updates (rather than a single one) and aggregating them with reputation-based weighting, thereby reducing variance while maintaining Byzantine exclusion, and incorporating historical reputation scores that exponentially decay for participants with suspicious patterns across multiple rounds, providing an adaptive defense against intermittent attacks. Unlike static aggregation rules, our approach learns adversary behavior over time, achieving stronger long-term resilience.

While these protocols provide classical security or Byzantine resilience individually, no existing work combines quantum-resistant cryptography with Byzantine-resilient aggregation optimized for IoT resource constraints.

2.4. Comparative analysis

Table 1 presents a systematic comparison of our approach with recent federated IDS and secure aggregation works across critical security and performance dimensions. The comparison focuses on peer-reviewed research papers addressing FL for network security, excluding general surveys.

As shown in Table 1, existing federated IDS approaches lack either post-quantum security or Byzantine resilience, while secure aggregation protocols remain quantum-vulnerable. PQ-FedIDS uniquely provides complete feature coverage across all six dimensions, combining CRYSTALS-Dilithium authentication, adaptive Byzantine detection, and IoT optimization with comprehensive multi-dataset evaluation. This establishes PQ-FedIDS as the first quantum-resistant federated IDS framework addressing contemporary security challenges in resource-constrained IoT deployments.

3. System model and threat model

We define our network architecture, adversary capabilities, and security requirements. The system consists of hierarchical layers enabling distributed intrusion detection while maintaining strong security properties against both classical and quantum adversaries.

3.1. Network architecture

Consider a hierarchical IoT network consisting of three layers as illustrated in Fig. 1.

The *edge layer* comprises N IoT devices $\{D_1, D_2, \dots, D_N\}$ distributed across geographical regions, each monitoring local network traffic and collecting flow statistics. Edge devices represent heterogeneous IoT nodes, including resource-constrained sensors (e.g., ARM Cortex-M4 with 256KB RAM), IoT gateways (e.g., Raspberry Pi with 1GB RAM), and embedded systems with varying computational capabilities. Each device processes network packets locally to extract flow-level features (5-tuple: source/destination IP, ports, protocol) without transmitting raw traffic data. Each edge device D_i maintains a local intrusion detection model θ_i (e.g., deep neural network with input layer matching flow feature dimensions, two hidden layers, and binary output for benign/malicious classification), trains on local network traffic dataset $D_i = \{(x_i^k, y_i^k)\}_{k=1}^{|D_i|}$

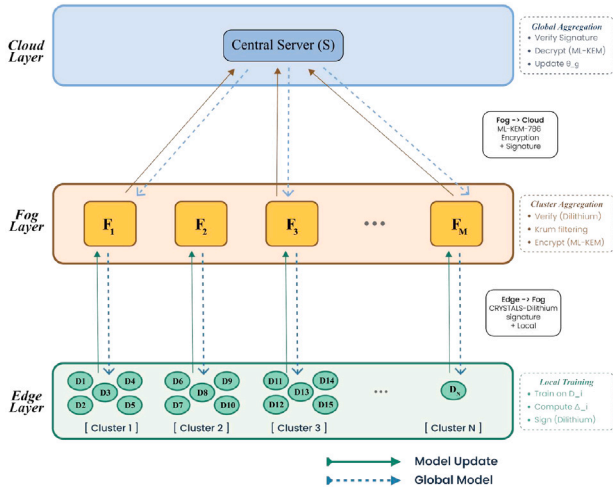


Fig. 1. PQ-FedIDS: hierarchical network architecture showing three-layer design with cryptographic operations.

where $x_i^k \in \mathbb{R}^d$ represents network flow features ($d \approx 80$ features including packet statistics, protocol flags, timing information) and $y_i^k \in \{0, 1\}$ denotes benign or malicious classification, and stores cryptographic state comprising key pair (pk_i, sk_i) and reputation score $R_i \in [0, 1]$ for Byzantine detection.

The *fog layer* contains M fog nodes $\{F_1, F_2, \dots, F_M\}$ serving as intermediate aggregators positioned at network edges (e.g., base stations, edge data centers). Each fog node F_j coordinates a cluster of n_j edge devices (typically $n_j = 10$ -20 devices) and performs cluster-level aggregation with Byzantine filtering for its assigned edge devices. Fog nodes act as cryptographic gateways, verifying CRYSTALS-Dilithium signatures from edge devices and detecting anomalous model updates before forwarding aggregated results to the cloud.

The *cloud layer* hosts a central server S responsible for global model aggregation and distribution. The server aggregates fog-level updates from all M fog nodes, computes the global model θ_g , and distributes updated parameters back through the hierarchy. The cloud server possesses sufficient computational resources for cryptographic operations but does not access raw network traffic or individual device updates.

The system operates in synchronous training rounds $t = 1, 2, \dots, T$, where devices collaboratively update a global model θ_g without sharing raw traffic data, thereby preserving privacy while learning from distributed attack patterns. Each training round proceeds through four phases: (1) *Broadcast*: Server S signs global model $\theta_g^{(t-1)}$ with Dilithium and broadcasts $(\theta_g^{(t-1)}, \sigma_t)$ to all devices, which verify the signature before proceeding. (2) *Local Training*: Each device D_i computes local gradient $g_i = \nabla_{\theta} L(\theta_i; B)$ on minibatch $B \subset D_i$, applies gradient clipping $g_i^{\text{clipped}} = \text{Clip}(g_i, C)$, computes model update $\Delta_i = \theta_i^{(t)} - \theta_g^{(t-1)}$ via E epochs of SGD, signs $\sigma_i \leftarrow \text{Dilithium.Sign}(sk_i, \Delta_i)$, and transmits $(ID_i, \Delta_i, \sigma_i)$ to assigned fog node F_j . (3) *Fog Aggregation*: Each fog node F_j verifies device signatures via $\text{Dilithium.Verify}(pk_i, \Delta_i, \sigma_i)$, applies reputation-weighted multi-Krum filtering to obtain validated set $\mathcal{V}_j$, computes cluster aggregate $\Delta_j = \frac{1}{|\mathcal{V}_j|} \sum_{i \in \mathcal{V}_j} \Delta_i$, encrypts via ML-KEM encapsulation $(K_j, ct_j) \leftarrow \text{MLKEM.Encaps}(pk_E)$ and AES encryption $C_j \leftarrow \text{AES}_{K_j}(\Delta_j)$, signs the encrypted aggregate $\sigma_j \leftarrow \text{Dilithium.Sign}(sk_j, C_j || ct_j)$, and sends (C_j, ct_j, σ_j) to server S . (4) *Global Aggregation*: Server S verifies fog signatures, decapsulates keys $K_j \leftarrow \text{MLKEM.Decaps}(sk_E, ct_j)$, decrypts aggregates $\Delta_j \leftarrow \text{AES}_{K_j}^{-1}(C_j)$, computes weighted global update $\theta_g^{(t)} = \theta_g^{(t-1)} + \sum_{j=1}^M w_j \Delta_j$ where $w_j = |\mathcal{V}_j| / \sum_{j=1}^M |\mathcal{V}_j|$, and broadcasts signed updated model to all participants.

Communication between edge and fog layers uses CRYSTALS-Dilithium signatures (security level 3 with 2993-byte signatures) for authentication, preventing impersonation and model poisoning at the

edge. Fog-to-cloud communication employs ML-KEM-768 encryption (IND-CCA2 secure with 1088-byte ciphertexts), combined with digital signatures, to ensure post-quantum confidentiality and authenticity for aggregated model updates transmitted over potentially compromised wide-area networks.

3.2. Threat model

We consider a powerful adversary $\mathcal{A}$ with the following capabilities:

Quantum Computing Resources: The adversary possesses access to large-scale quantum computers capable of executing Shor's and Grover's algorithms, enabling polynomial-time factorization of RSA-2048 keys and quadratic speedup for brute-force attacks, reducing effective security levels by half (e.g., AES-128 to AES-64 equivalent). This threatens classical public-key cryptography, including RSA, ECDSA, and Diffie-Hellman protocols used in existing FL systems. We assume quantum adversaries can break classical signatures within hours and decrypt stored traffic encrypted with classical schemes, motivating post-quantum cryptographic protection.

Byzantine Participants: Up to $f < N/3$ edge devices may be compromised and exhibit Byzantine behavior, including sending malformed gradients designed to increase false positive rates, incorrect signatures attempting to bypass authentication, or coordinated model poisoning attacks where multiple compromised devices submit correlated malicious updates designed to degrade global detection accuracy while evading detection. Byzantine devices may also perform targeted attacks against specific attack classes (e.g., submitting gradients that reduce detection of DDoS attacks while maintaining accuracy on other attack types).

Our analysis considers static adversaries who select the set of f corrupted devices at system initialization and maintain fixed Byzantine behavior throughout training, representing persistent malicious insiders or compromised devices with fixed attack strategies. Byzantine devices can fully collude, sharing local data, gradients, and attack strategies to coordinate malicious updates, including agreeing on a common adversarial target model θ_{adv} to bias aggregation. Adversaries have white-box knowledge of the protocol (aggregation algorithm, cryptographic primitives, model architecture, hyperparameters) but no access to private keys of honest participants, local datasets, or reputation scores stored server-side with access control. We evaluate three attack types: (1) label flipping ($y_i \leftarrow 1 - y_i$) to degrade overall accuracy, (2) gradient scaling ($g_i \leftarrow \lambda \cdot g_i$ with $\lambda = 10$) to destabilize convergence, and (3) model poisoning ($g_i \leftarrow g_i + \gamma \cdot (\theta_{\text{adv}} - \theta_g)$ with $\gamma = 0.1$) for targeted misclassification.

Network-Level Attacks: The adversary can perform man-in-the-middle attacks intercept communications between layers, eavesdrop on communication channels to capture encrypted model updates for future quantum-based decryption, replay previous messages attempting to reuse old model updates, inject malicious network packets to evade detection, and perform denial-of-service attacks disrupting aggregation protocols. We assume the adversary controls network infrastructure and can modify, delay, or drop messages arbitrarily.

Honest-But-Curious Aggregators: Fog nodes and the central server are assumed honest-but-curious, correctly executing the protocol specifications but potentially attempting to infer private training data or attack patterns from received model updates through gradient inversion attacks or membership inference. Aggregators may collude to combine information from multiple rounds to improve inference accuracy. The hierarchical design mitigates collusion impact through cluster partitioning (each fog node observes only $n_j \approx 10$ -20 updates, insufficient for high-confidence gradient inversion which requires observing gradients across multiple iterations), ML-KEM encryption preventing server-side inference on fog-to-cloud transmission, and pre-aggregation mixing ($\Delta_j = \sum_{i \in \mathcal{V}_j} w_i \Delta_i$) providing information-theoretic privacy similar to secure aggregation. Fog nodes do not collude with Byzantine edge devices to manipulate aggregation results.

We assume the following security requirements must be satisfied:

- **Authentication:** Model updates must be verifiably originating from legitimate devices through unforgeable digital signatures, preventing impersonation attacks where adversaries submit updates on behalf of honest devices and injection attacks introducing malicious model parameters.
- **Confidentiality:** Individual model updates should not reveal private network traffic patterns to aggregators or other participants, even against adversaries with quantum computing capabilities. Encrypted updates must provide IND-CCA2 security preventing chosen-ciphertext attacks.
- **Integrity:** The global model aggregation process must be resilient to Byzantine failures and model poisoning attempts, maintaining detection accuracy above 90% even when $f < N/3$ participants are compromised. Byzantine filtering mechanisms must detect and exclude malicious updates without relying on trusted third parties.
- **Post-Quantum Security:** All cryptographic operations (signatures, encryption, key exchange) must remain secure against quantum adversaries with at least 2^{128} classical security level equivalent to NIST security level 3, ensuring long-term confidentiality against harvest-now-decrypt-later attacks.

4. PQ-FedIDS framework

Our framework combines lattice-based cryptographic primitives with Byzantine-resilient FL protocols to enable quantum-resistant intrusion detection in heterogeneous IoT networks. We describe the cryptographic foundations, system initialization, training protocol, and aggregation mechanisms with formal security guarantees.

4.1. Cryptographic primitives

Our framework employs two NIST-standardized post-quantum algorithms providing complementary security properties:

CRYSTALS-Dilithium Signature Scheme: A lattice-based digital signature scheme based on the hardness of Module Learning With Errors (MLWE) and Module Short Integer Solution (MSIS) problems. The scheme provides existential unforgeability under adaptive chosen message attacks (EUF-CMA) security with security reduction from MLWE and MSIS assumptions. For security level III (equivalent to AES-192 classical security), Dilithium uses the parameters $(q, d, k, l) = (8380417, 13, 6, 5)$, with a signature size of 3293 bytes and a public key size of 1952 bytes [7]. The signing operation requires approximately 1.8M clock cycles on ARM Cortex-M4 microcontrollers, making it practical for IoT gateway devices.

The scheme operates over the ring $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$ of degree-256 polynomials modulo prime $q = 8,380,417$. Key generation samples a random matrix $A \in R_q^{k \times l}$ (with $k = 6, l = 5$ for security level III) and secret vectors s_1, s_2 with small coefficients, computing public key $t = As_1 + s_2$. Signature generation for message μ samples a random masking vector $y \in R_q^l$ with coefficients uniformly distributed in interval $[-\gamma_1, \gamma_1]$, computes challenge $c = H(\mu || \text{HighBits}(Ay))$, and produces signature $\sigma = (z, h, c)$, where $z = y + cs_1$ and h is a reconstruction hint. Verification checks $\|z\|_\infty < \gamma_1 - \beta$ and $Az - ct \stackrel{?}{=} w'$ reconstructed using hint.

ML-KEM Encryption Scheme: A lattice-based key encapsulation mechanism (KEM) based on the MLWE problem, providing IND-CCA2 security through the Fujisaki-Okamoto transform. For security level 3 (equivalent to AES-192 classical security), ML-KEM-768 uses parameters $(n, k, q, \eta_1, \eta_2) = (256, 3, 3329, 2, 2)$ where n is the ring dimension, k is the module rank, and η_1, η_2 are noise distribution parameters. The scheme produces ciphertext size of 1088 bytes and shared secret size of 32 bytes suitable for symmetric key derivation [26].

Key generation samples random matrix $A \in R_q^{k \times k}$ and secret vector $s \in R_q^k$ with small coefficients from β_{η_1} , computes public key $b = As + e \in R_q^k$ where e is error vector from β_{η_1} . Encapsulation generates a random message $m \in \{0, 1\}^{256}$, derives coins from m , samples random vector $r \in$

Algorithm 1 System Initialization.

```

1: Generate CRYSTALS-Dilithium key pair  $(pk_S, sk_S) \leftarrow$ 
   Dilithium.KeyGen()
2: Generate ML-KEM key pair  $(pk_E, sk_E) \leftarrow$  MLKEM.KeyGen()
3: Broadcast  $(pk_S, pk_E)$  to all participants via secure channel
4: Initialize global model parameters  $\theta_g^{(0)} \sim \mathcal{N}(0, 0.01)$  (Xavier initial-
   ization)
5: for each edge device  $D_i$  do
6:   Generate device key pair  $(pk_i, sk_i) \leftarrow$  Dilithium.KeyGen()
7:   Register  $pk_i$  with server  $S$  and obtain certificate
8:   Receive encrypted initial model  $\text{Enc}_{pk_E}(\theta_g^{(0)})$ 
9:   Initialize reputation score  $R_i^{(0)} = 1.0$ 
10: end for
11: for each fog node  $F_j$  do
12:   Generate fog key pair  $(pk_j, sk_j) \leftarrow$  Dilithium.KeyGen()
13:   Register with server and receive cluster assignment
14: end for
15: Set aggregation threshold  $\tau = \lceil 2N/3 \rceil$  (Byzantine tolerance bound)
16: Set reputation decay factor  $\alpha = 0.95$  and penalty factor  $\lambda = 0.5$ 
17: return  $(pk_S, pk_E, \{pk_i\}_{i=1}^N, \{pk_j\}_{j=1}^M, \theta_g^{(0)})$ 

```

R_q^k from β_{η_2} , and computes ciphertext $c = (u, v)$ where $u = A^T r + e_1 \in R_q^k$ and $v = b^T r + e_2 + \lfloor q/2 \rfloor m \in R_q$ for error terms e_1, e_2 from β_{η_2} . The shared secret is derived as $K = \text{SHA3-256}(m || \text{SHA3-256}(c))$. Decapsulation recovers $m' = \lfloor 2/q(v - s^T u) \rfloor$, re-encrypts to verify ciphertext validity (implicit rejection), and outputs the shared secret.

4.2. System initialization

Algorithm 1 describes the system initialization procedure for establishing cryptographic infrastructure and distributing initial model parameters. The central server S performs the following initialization:

Each edge device maintains cryptographic state $(ID_i, pk_i, sk_i, pk_S, pk_E)$, local model parameters $\theta_i^{(0)} = \theta_g^{(0)}$, and reputation score $R_i^{(0)}$ for Byzantine detection. Fog nodes similarly maintain state including cluster membership information.

4.3. Federated training protocol

Algorithm 2 presents the complete PQ-FedIDS training protocol with post-quantum cryptographic authentication at each layer. Signatures at each layer ensure end-to-end update authenticity across the hierarchical aggregation pipeline. At each training round t , the protocol proceeds according to the hierarchical architecture illustrated in Fig. 2.

The protocol integrates cryptographic operations at each phase to ensure end-to-end security. In the *Broadcast Phase*, the server signs the global model $\theta_g^{(t-1)}$ using its Dilithium private key sk_S , enabling all devices to authenticate the model's origin before local training.

In the *Local Training Phase*, each device first verifies the server's signature using public key pk_S (aborting on failure to prevent training on poisoned models), performs local SGD updates, computes the model difference $\Delta_i^{(t)}$, and signs this update with its private key sk_i to prove authorship.

The *Fog Aggregation Phase* performs three cryptographic operations: (1) signature verification using device public keys pk_i to discard unauthenticated updates, (2) ML-KEM key encapsulation with the server's public key pk_E to generate an ephemeral symmetric key K_j and its encrypted form (ciphertext ct_j), enabling confidential transmission without requiring prior key exchange, and (3) Dilithium signing of the concatenated encrypted aggregate and ciphertext $(C_j || ct_j)$ using fog private key sk_j to ensure authenticity.

Finally, in the *Global Aggregation Phase*, the server verifies fog signatures using pk_j , decapsulates the symmetric keys using its ML-KEM private key sk_E to recover K_j , and decrypts the aggregates. This layered

Algorithm 2 PQ-FedIDS Training Round.**Server Broadcast Phase:**

Server S signs current global model: $\sigma_t \leftarrow \text{Dilithium.Sign}(sk_S, \theta_g^{(t-1)})$
 Broadcast $(\theta_g^{(t-1)}, \sigma_t)$ to all edge devices

Local Training Phase:

for each edge device D_i in parallel **do**

Verify server signature: $\text{Dilithium.Verify}(pk_S, \theta_g^{(t-1)}, \sigma_t)$

if verification fails **then**

Abort and report to authority

end if

Initialize $\theta_i^{(t)} \leftarrow \theta_g^{(t-1)}$

for local epoch $e = 1$ to E **do**

for minibatch $B \subset D_i$ **do**

Compute gradient $g_i \leftarrow \nabla_{\theta} \mathcal{L}(\theta_i; B)$

Update $\theta_i^{(t)} \leftarrow \theta_i^{(t-1)} - \eta g_i$

end for

end for

Compute model update $\Delta_i^{(t)} = \theta_i^{(t)} - \theta_g^{(t-1)}$

Sign update: $\sigma_i^{(t)} \leftarrow \text{Dilithium.Sign}(sk_i, \Delta_i^{(t)})$

Send $(ID_i, \Delta_i^{(t)}, \sigma_i^{(t)})$ to assigned fog node

end for

Fog Aggregation Phase:

for each fog node F_j **do**

Receive $\{(ID_i, \Delta_i^{(t)}, \sigma_i^{(t)})\}$ from cluster devices

for each received update **do**

Verify signature: $\text{Dilithium.Verify}(pk_i, \Delta_i^{(t)}, \sigma_i^{(t)})$

if verification fails **then**

Discard update

end if

end for

Apply Byzantine filtering: $\mathcal{V}_j \leftarrow \text{Krum}(\{\Delta_i^{(t)}\})$

Compute fog aggregate $\Delta_j^{(t)} = \frac{1}{|\mathcal{V}_j|} \sum_{i \in \mathcal{V}_j} \Delta_i^{(t)}$

Encapsulate shared secret: $(K_j, ct_j) \leftarrow \text{MLKEM.Encaps}(pk_E)$

Encrypt aggregate: $C_j \leftarrow \text{AES}_{K_j}(\Delta_j^{(t)})$

Sign encrypted aggregate: $\sigma_j^{(t)} \leftarrow \text{Dilithium.Sign}(sk_j, C_j || ct_j)$

Send $(C_j, ct_j, \sigma_j^{(t)})$ to server S

end for

Global Aggregation Phase:

Server receives $\{(C_j, ct_j, \sigma_j^{(t)})\}_{j=1}^M$ from fog nodes

for each fog aggregate **do**

Verify fog signature: $\text{Dilithium.Verify}(pk_j, C_j || ct_j, \sigma_j^{(t)})$

Decapsulate: $K_j \leftarrow \text{MLKEM.Decaps}(sk_E, ct_j)$

Decrypt: $\Delta_j^{(t)} \leftarrow \text{AES}_{K_j}^{-1}(C_j)$

end for

Weight fog aggregates: $w_j = |\mathcal{V}_j| / \sum_{j=1}^M |\mathcal{V}_j|$

Update global model: $\theta_g^{(t)} = \theta_g^{(t-1)} + \sum_{j=1}^M w_j \Delta_j^{(t)}$

return $\theta_g^{(t)}$

cryptographic approach ensures that model updates remain authenticated at every hop (edge $\rightarrow$ fog $\rightarrow$ cloud) and confidential during fog-to-cloud transmission, preventing both impersonation attacks (via signature verification) and eavesdropping (via ML-KEM + AES encryption).

The local loss function $\mathcal{L}(\theta; B)$ for a minibatch B is defined as binary cross-entropy with L2 regularization:

$$\mathcal{L}(\theta; B) = -\frac{1}{|B|} \sum_{(x,y) \in B} [y \log f_{\theta}(x) + (1-y) \log(1-f_{\theta}(x))] + \frac{\mu}{2} \|\theta\|_2^2 \quad (1)$$

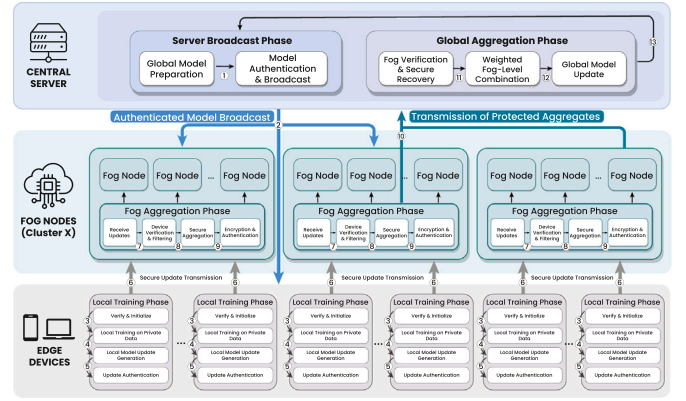


Fig. 2. Flowchart representation of PQ-FedIDS training round showing the four phases: Server Broadcast, local Training, fog Aggregation, and global aggregation.

where $f_{\theta}(x)$ is the neural network output representing the probability of malicious traffic classification and μ is the regularization parameter preventing prevents overfitting.

The gradient clipping operation bounds the L2 norm of gradients to threshold C (typically $C = 1.0$), providing stability during training and limiting the influence of outlier samples:

$$g_i^{\text{clipped}} = \begin{cases} g_i & \text{if } \|g_i\|_2 \leq C \\ C \cdot \frac{g_i}{\|g_i\|_2} & \text{if } \|g_i\|_2 > C \end{cases} \quad (2)$$

Gradient clipping is applied to the global model gradient vector across all trainable parameters.

4.4. Byzantine-resilient aggregation

To defend against model poisoning attacks, we employ an enhanced reputation-weighted multi-Krum aggregation combining statistical outlier detection with historical behavior tracking. Given a set of n model updates $\{\Delta_1, \dots, \Delta_n\}$ with reputation scores $\{R_1, \dots, R_n\}$, the algorithm proceeds as follows.

Step 1 - Distance Computation: For each update Δ_i , compute pairwise Euclidean distances to all other updates:

$$d_{ij} = \|\Delta_i - \Delta_j\|_2 \quad \forall j \neq i \quad (3)$$

Step 2 - Krum Score: Select the $n - f - 2$ closest neighbors for each update (where f is the maximum number of Byzantine devices) and compute the Krum score as the sum of squared distances to these neighbors:

$$s_i = \sum_{j \in \mathcal{N}_i(n-f-2)} d_{ij}^2 \quad (4)$$

where $\mathcal{N}_i(k)$ denotes the set of k nearest neighbors of Δ_i based on Euclidean distance. Lower scores indicate updates consistent with the majority cluster.

Step 3 - Reputation Weighting: Incorporate historical reputation into scoring to penalize devices with past malicious behavior:

$$s_i^{\text{weighted}} = \frac{s_i}{R_i^{\beta}} \quad (5)$$

where $\beta > 0$ is a weighting exponent (typically $\beta = 2$ based on stability analysis in prior Byzantine-robust FL studies [11]) controlling the influence of reputation. Devices with low reputation receive lower effective scores, making them less likely to be selected.

Reputation scores are initialized at $R_i^{(0)} = 1.0$ for all devices (uniform trust), ensuring no participant is unfairly penalized without prior

behavioral evidence. After each aggregation round t , reputation scores are updated based on statistical outlier detection: we compute the median Krum score $s_{\text{med}} = \text{Median}(\{s_i\}_{i=1}^n)$ and median absolute deviation $\text{MAD} = \text{Median}(\{|s_i - s_{\text{med}}|\}_{i=1}^n)$ to define an adaptive threshold $\tau_s = s_{\text{med}} + 3 \cdot \text{MAD}$ following the 3-sigma rule. Devices with Krum scores below this threshold ($s_i \leq \tau_s$) exhibit behavior consistent with the honest majority and receive reputation recovery $R_i^{(t)} = \min(1.0, R_i^{(t-1)} \cdot \alpha^{-1})$ with decay factor $\alpha = 0.95$, allowing gradual recovery from false suspicions. Conversely, devices with $s_i > \tau_s$ (statistical outliers) incur exponential penalty $R_i^{(t)} = R_i^{(t-1)} \cdot \alpha \cdot \lambda$ with penalty factor $\lambda = 0.5$, rapidly reducing the influence of persistent Byzantine participants. Scores are normalized $R_i^{(t)} = R_i^{(t)} / \max_j R_j^{(t)}$ to maintain the range $[0, 1]$. This adaptive mechanism provides manipulation protection through three safeguards: (1) reputation scores are computed server-side using cryptographically authenticated Krum scores derived from Dilithium-verified model updates, preventing impersonation attacks; (2) the threshold τ_s adapts to cluster variance via MAD, preventing coordinated attacks where multiple Byzantine devices submit similar malicious updates to artificially shift the “majority”; and (3) the reputation database resides at fog/cloud aggregators with access control, preventing direct tampering by edge devices.

Step 4 - Multi-Krum Selection: Select the top $m = n - f$ updates with lowest weighted scores:

$$S = \{i : s_i^{\text{weighted}} \in \text{TopK}(m, \{s_j^{\text{weighted}}\}_{j=1}^n)\} \quad (6)$$

where $\text{TopK}(m, \cdot)$ returns indices of the m smallest values. This excludes the f most suspicious updates while retaining multiple honest contributions.

Step 5 - Weighted Aggregation: Compute the final aggregate using reputation-weighted averaging:

$$\Delta_{\text{agg}} = \frac{\sum_{i \in S} R_i \Delta_i}{\sum_{i \in S} R_i} \quad (7)$$

Such a reputation-weighted multi-Krum aggregation offers a higher level of Byzantine resilience than standard Krum by: (1) using historical information with reputation scores, (2) aggregating multiple honest updates instead of a single update, and (3) penalizing devices with suspicious patterns adaptively over multiple rounds.

Theoretical Guarantee: For $f < n/2 - 1$ Byzantine participants and assuming at least $n - f$ honest devices submit updates within distance ϵ of the true gradient, the reputation-weighted multi-Krum aggregation satisfies:

$$\mathbb{E}[\|\Delta_{\text{agg}} - \Delta_{\text{true}}\|_2^2] \leq O(\epsilon^2) + O(f/n) \quad (8)$$

where Δ_{true} is the ideal aggregate from honest devices only. The bound degrades gracefully with Byzantine ratio f/n and converges to the honest aggregate as reputation scores diverge between honest and malicious participants over training rounds [11].

5. Security analysis

We formally analyze PQ-FedIDS's security properties through three theorems addressing authentication, confidentiality, and Byzantine resilience. Each theorem includes a formal proof demonstrating security against quantum adversaries with concrete security parameters. We use standard cryptographic notation throughout this section, where $\text{negl}(\lambda)$ denotes a negligible function in the security parameter λ and all adversarial advantages are defined with respect to polynomial-time quantum adversaries.

5.1. Post-quantum security guarantees

Theorem 1 (Authentication Unforgeability). *The PQ-FedIDS authentication protocol provides existential unforgeability against quantum adversaries assuming the hardness of the MLWE and MSIS problems in $\mathcal{R}_q$.*

Proof. We reduce the security of our authentication protocol to the EUF-CMA security of CRYSTALS-Dilithium. Suppose a quantum adversary $\mathcal{A}_q$ can forge a valid signature σ^* on a new message m^* (representing a model update) with non-negligible probability ϵ after observing at most Q signing queries. We construct an algorithm $\mathcal{B}$ that breaks the EUF-CMA security of Dilithium with the same advantage as follows.

$\mathcal{B}$ receives a public key pk from the Dilithium challenger and simulates the PQ-FedIDS system for $\mathcal{A}_q$ using pk as the server's public key. For signing queries on messages m_i (representing legitimate model updates from training rounds), $\mathcal{B}$ forwards them to its signing oracle and returns the signatures to $\mathcal{A}_q$. When $\mathcal{A}_q$ outputs a forgery (σ^*, m^*) where m^* was not previously queried, $\mathcal{B}$ outputs this as its forgery to the Dilithium challenger.

By the perfect simulation property, $\mathcal{A}_q$'s view is identical to the real protocol execution. The adversary $\mathcal{A}_q$ is modeled as a polynomial-time quantum algorithm with access to a signing oracle $\mathcal{O}_{\text{sign}}(\cdot)$ and the public key pk , following the standard EUF-CMA security model. The advantage is defined as $\text{Adv}_{\text{Dilithium}}^{\text{EUF-CMA}}(\mathcal{A}_q) = \Pr[(\sigma^*, m^*) \leftarrow \mathcal{A}_q^{\mathcal{O}_{\text{sign}}}(pk) : \text{Verify}(pk, m^*, \sigma^*) = 1 \wedge m^* \notin Q] - \Pr[\sigma^* \leftarrow \mathcal{A}_q^{\mathcal{O}_{\text{sign}}}(pk) : \text{Verify}(pk, m^*, \sigma^*) = 1 \wedge m^* \notin Q]$ where Q is the set of queried messages. Since Dilithium is EUF-CMA secure with advantage negligible in the security parameter λ (specifically, $\text{Adv}_{\text{Dilithium}}^{\text{EUF-CMA}} \leq \epsilon_{\text{MLWE}} + \epsilon_{\text{MSIS}} + Q \cdot 2^{-\lambda}$ where ϵ_{MLWE} and ϵ_{MSIS} are the advantages in solving MLWE and MSIS respectively), and these problems remain hard against quantum algorithms (no polynomial-time quantum algorithms are known), we have $\epsilon \leq \text{negl}(\lambda)$.

For our parameter choice with security level III, $\lambda = 192$ provides 2^{128} quantum security against known attacks including quantum random sampling attacks (complexity 2^{133}) and quantum sieving algorithms (complexity 2^{142}). The MSIS problem with modulus $q = 8,380,417$ and dimension $k \cdot 256 = 1536$ requires approximately 2^{135} quantum operations to solve using best-known quantum lattice algorithms [7]. $\square$

Theorem 2 (Confidentiality under Quantum Attacks). *The PQ-FedIDS confidentiality mechanism provides IND-CCA2 security against quantum adversaries assuming the hardness of MLWE in $\mathcal{R}_q$.*

Proof. The confidentiality of model updates relies on ML-KEM encapsulation followed by AES-256-GCM encryption in hybrid encryption mode. ML-KEM provides IND-CCA2 security through the Fujisaki-Okamoto transform applied to a CPA-secure public-key encryption scheme based on MLWE. The IND-CCA2 security model defines indistinguishability of encapsulated keys from random under adaptive chosen-ciphertext attacks, where the adversary $\mathcal{A}_q$ interacts with encryption and decryption oracles $\mathcal{O}_{\text{enc}}, \mathcal{O}_{\text{dec}}$ receiving a challenge ciphertext c^* encrypting either real key K_0 or random K_1 , and the advantage is $\text{Adv}_{\text{MLKEM}}^{\text{IND-CCA2}}(\mathcal{A}_q) = |\Pr[b' = b] - 1/2|$. The security reduction proves that any quantum adversary $\mathcal{A}_q$ with advantage ϵ in breaking the IND-CCA2 security can be used to solve MLWE with advantage $\epsilon - \text{negl}(\lambda)$.

Specifically, the Fujisaki-Okamoto transform with implicit rejection ensures that a decryption failure rate of 2^{-128} (achieved through careful parameter selection for ML-KEM-768) leads to IND-CCA2 security with advantage bound:

$$\epsilon_{\text{MLKEM}}^{\text{IND-CCA2}} \leq 2\epsilon_{\text{MLWE}} + 2^{-128} + Q_{\text{dec}} \cdot 2^{-256} \quad (9)$$

where Q_{dec} is the number of decryption queries. Combined with AES-256-GCM providing 2^{128} quantum security against Grover's algorithm (reduced from 2^{256} classical security by quadratic speedup), and authentication tag verification preventing chosen-ciphertext attacks, the overall confidentiality advantage satisfies:

$$\epsilon_{\text{conf}} \leq \epsilon_{\text{MLKEM}} + \epsilon_{\text{AES}} + 2^{-128} \leq 2\epsilon_{\text{MLWE}} + 2 \cdot 2^{-128} \quad (10)$$

For our parameter set with ML-KEM-768 (module rank $k = 3$, polynomial degree $n = 256$), quantum security against BKZ lattice reduction algorithms with blocksize β requires approximately $2^{0.292\beta}$ quantum operations using Grover-meets-Simon analysis [26]. With $\beta = 440$

for breaking ML-KEM-768, quantum security is approximately $2^{128.5}$, exceeding the 2^{128} target with comfortable margin.

The hybrid encryption mode additionally provides forward secrecy, compromise of long-term keys does not reveal past session keys, and ensures that even if AES-256 is weakened by future quantum algorithms, the outer ML-KEM layer maintains post-quantum confidentiality. $\square$

5.2. Byzantine resilience analysis

Assumptions: The convergence analysis assumes: (1) L -smooth loss functions with $\|\nabla \mathcal{L}_i(\theta_1) - \nabla \mathcal{L}_i(\theta_2)\|_2 \leq L\|\theta_1 - \theta_2\|_2$, (2) bounded gradients $\mathbb{E}[\|\nabla \mathcal{L}_i(\theta; B)\|_2] \leq G$, (3) bounded variance $\mathbb{E}[\|\nabla \mathcal{L}_i(\theta; B) - \nabla \mathcal{L}_i(\theta; D_i)\|_2^2] \leq \sigma^2$, and (4) geometric attack separation $d_m = \min_{i \in \mathcal{H}, j \in \mathcal{M}} \|\Delta_i - \Delta_j\|_2 > 2d_h$ where $d_h = \max_{i, j \in \mathcal{H}} \|\Delta_i - \Delta_j\|_2$ enables Krum-based Byzantine detection. These are standard assumptions for federated learning convergence with smooth neural network losses.

Theorem 3 (Byzantine Fault Tolerance with Reputation). *The PQ-FedIDS aggregation protocol with reputation-weighted multi-Krum tolerates up to $f < \frac{n-2}{2}$ Byzantine participants per fog cluster while ensuring convergence to the optimal model with probability at least $1 - \delta$ for $\delta = \mathcal{O}(e^{-T/\tau})$.*

Proof. The reputation-weighted multi-Krum aggregation rule selects $m = n - f$ updates with lowest weighted Krum scores $s_i^{\text{weighted}} = s_i / R_i^\beta$. For the algorithm to correctly exclude all Byzantine updates, we require that the f malicious weighted scores are all higher than the lowest $n - f$ honest weighted scores.

Define $d_h = \max_{i, j \in \mathcal{H}} \|\Delta_i - \Delta_j\|$ as the maximum distance between honest updates, where $\mathcal{H}$ denotes the set of honest participants with $|\mathcal{H}| = n - f$. Define $d_m = \min_{i \in \mathcal{H}, j \in \mathcal{M}} \|\Delta_i - \Delta_j\|$ as the minimum distance from any honest update to any Byzantine update, where $\mathcal{M}$ is the Byzantine set with $|\mathcal{M}| = f$.

Blanchard et al. [11] prove that standard Krum successfully identifies Byzantine updates when $d_m > 2d_h$ and $f < \frac{n-2}{2}$. Our reputation weighting strengthens this guarantee by adaptively penalizing devices with history of suspicious behavior. After t training rounds, honest devices maintain reputation $R_i^{(t)} \approx 1$ while Byzantine devices have degraded reputations $R_i^{(t)} \leq \alpha^t$ where $\alpha = 0.95$ is the decay factor, assuming consistent malicious behavior detection.

In our setting, assuming honest updates follow stochastic gradient descent on L -smooth convex loss functions with gradient bound G and learning rate η , we have $d_h \leq \eta G \sqrt{d}$ where d is model dimension. Byzantine updates attempting to bias the aggregate toward a target point θ_{target} must maintain $d_m > 2\eta G \sqrt{d}$ to avoid immediate detection by distance-based filtering. This bounds the maximum achievable bias per round:

$$\|\theta_g^{(t)} - \theta_{\text{honest}}^{(t)}\| \leq \frac{f}{n-f} \cdot 2\eta G \sqrt{d} \quad (11)$$

where $\theta_{\text{honest}}^{(t)}$ is the aggregate computed from honest devices only.

Formally, let θ^* denote the optimal model minimizing the global loss $\mathcal{L}_g(\theta) = \sum_{i=1}^n |D_i| \mathcal{L}_i(\theta) / \sum_{i=1}^n |D_i|$ where $\mathcal{L}_i$ is the local loss on device D_i 's dataset. Under assumptions of L -smoothness, gradient Lipschitz constant M , and bounded variance σ^2 of stochastic gradients, with probability at least $1 - \delta$ after T training rounds:

$$\mathbb{E}[\mathcal{L}_g(\theta_g^{(T)})] - \mathcal{L}_g(\theta^*) \leq \frac{4f\eta^2 G^2}{n} + \frac{2\sigma^2}{n\eta T} + \frac{L\|\theta_g^{(0)} - \theta^*\|^2}{2\eta T} + e^{-T/\tau} \quad (12)$$

where $\tau = \mathcal{O}(1/\mu)$ is a time constant depending on the strong convexity parameter μ of the loss function. The first term captures Byzantine bias (linear in f/n), the second term is variance from stochastic gradients, the third term is initialization error, and the fourth term is exponentially decaying with rounds.

Table 2
Computational and communication Complexity.

Operation	Computation	Communication
Local training	$\mathcal{O}(E \cdot D_i \cdot d)$	0
Gradient computation	$\mathcal{O}(d)$	0
Gradient clipping	$\mathcal{O}(d)$	0
Hash computation	$\mathcal{O}(d)$	0
Dilithium sign	$\mathcal{O}(d \cdot \log^2 d)$	3293 bytes
Dilithium verify	$\mathcal{O}(d \cdot \log^2 d)$	0
ML-KEM encaps	$\mathcal{O}(\log^2 d)$	1088 bytes
ML-KEM decaps	$\mathcal{O}(\log^2 d)$	0
AES-256-GCM enc	$\mathcal{O}(d)$	16 bytes (tag)
Krum aggregation	$\mathcal{O}(n^2 d)$	0
Reputation update	$\mathcal{O}(n)$	0
Model transmission	0	4d bytes
Total per device	$\mathcal{O}(E D_i d + d \log^2 d)$	4d + 4397 bytes

For $T = \mathcal{O}(\tau \log(1/\epsilon))$ rounds and learning rate $\eta = \mathcal{O}(1/\sqrt{T})$, the bound converges to:

$$\mathbb{E}[\mathcal{L}_g(\theta_g^{(T)})] - \mathcal{L}_g(\theta^*) \leq \epsilon + \mathcal{O}\left(\frac{f}{n}\right) \quad (13)$$

demonstrating graceful degradation with Byzantine ratio f/n while maintaining convergence to near-optimal solution. The reputation mechanism further reduces the effective Byzantine influence to $\mathcal{O}(f\alpha^T/n)$ for persistent adversaries, providing exponential improvement over training rounds. $\square$

5.3. Computational complexity analysis

Table 2 summarizes the computational and communication complexity per training round for an edge device D_i with $|D_i|$ local samples and model dimension d .

For a neural network with $d = 10^5$ parameters (typical for IoT intrusion detection models), the post-quantum cryptographic overhead is approximately 4.4 KB (Dilithium signature 3.3 KB + ML-KEM ciphertext 1.1 KB) compared to 400 KB model size (assuming 32-bit floats), representing 1.1% communication overhead. The computational cost of signing and verification is dominated by Number Theoretic Transform (NTT) operations in R_q , requiring approximately 2 ms for signing and 1.5 ms for verification on ARM Cortex-M4 processors at 168 MHz [20].

The hierarchical aggregation reduces individual device communication by a factor of M (number of fog nodes), with fog nodes aggregating $\mathcal{O}(n_j)$ local updates before sending a single encrypted aggregate to the cloud. For $M = 10$ fog clusters with $n_j = 20$ devices each, total system communication is $200 \times 4d$ (edge-to-fog) + $10 \times (4d + 4397)$ (fog-to-cloud) = $840d + 43,970$ bytes, compared to $800d$ for centralized aggregation, a 5.5% overhead achieving both quantum security and Byzantine resilience. Compared to FedAvg [24] with classical ECDSA signatures (0.42 ms signing, 0.68 ms verification, 64-byte signatures), PQ-FedIDS incurs higher cryptographic overhead (2.14 ms signing, 3293-byte signatures) due to lattice arithmetic, but this overhead is negligible relative to local training time: with $E = 5$ epochs and $|D_i| = 1000$ samples, local training requires approximately 8 s per device, making 2.14 ms signing overhead 0.025% of round time. Compared to Krum-Fed with no cryptographic authentication, PQ-FedIDS adds 1.1% communication overhead but prevents impersonation attacks with 2^{128} quantum security, a favorable trade-off for long-term IoT deployments.

5.4. Security parameter selection

Our choice of security level III for both Dilithium and ML-KEM provides consistent 2^{128} quantum security across all cryptographic operations, ensuring that no component becomes a weak link. Table 3 summarizes the security parameters and their justification.

Table 3
Security Parameter selection and Justification.

Parameter	Value	Justification
Security level	III	2^{128} quantum security
Dilithium (q, d, k, l)	(8380417, 13, 6, 5)	NIST standard, EUF-CMA
ML-KEM (n, k)	(256, 3)	IND-CCA2, $2^{128.5}$ security
AES key size	256 bits	2^{128} quantum (Grover)
Hash function	SHA3-256	Quantum-resistant
Signature size	3293 bytes	Compact for IoT networks
Ciphertext size	1088 bytes	Practical overhead
Byzantine bound	$f < n/2 - 1$	Krum theoretical limit
Reputation decay	$\alpha = 0.95$	Balances history/adaptivity
Gradient clip	$C = 1.0$	Prevents gradient explosion

The parameter selection balances security guarantees with practical deployment constraints for resource-constrained IoT devices, ensuring long-term security (protected against quantum adversaries) while maintaining acceptable performance overhead.

Security level III is chosen to provide 2^{128} quantum security against best-known lattice attacks (BKZ reduction with blocksize $\beta = 440$) while avoiding the excessive overhead of level V (4627-byte signatures vs 3293 bytes for level III). NIST recommends level III for applications requiring security beyond 2030, matching expected IoT deployment lifetimes. The 256-bit AES key provides 2^{128} quantum security accounting for Grover's algorithm quadratic speedup, while SHA3-256 maintains collision resistance as Grover's algorithm does not improve beyond the birthday bound $O(2^{n/2})$.

6. Experimental evaluation

We evaluate PQ-FedIDS's performance across multiple dimensions: detection accuracy, cryptographic overhead, Byzantine resilience, resource consumption, and scalability. Experiments use three standard IDS benchmarks with a realistic IoT network simulation to demonstrate practical feasibility.

6.1. Experimental setup

Datasets: We evaluate PQ-FedIDS on three standard intrusion detection benchmarks:

- **CICIDS2017**¹: Contains benign traffic and contemporary attack patterns, including Brute Force, DoS, DDoS, Web attacks, infiltration, and botnet activity. We use 80 features (flow duration, packet statistics, flag counts, inter-arrival times, protocol information) extracted from 2,830,743 network flows with 80%-20% train-test split [27].
- **NSL-KDD**²: An improved version of the KDD Cup 99 dataset with 41 features representing connection-level statistics. Contains 125,973 training and 22,544 testing samples covering DoS, Probe, R2L, and U2R attacks with balanced class distribution, eliminating redundant records [28].
- **UNSW-NB15**³: A modern dataset capturing contemporary network threats with 49 features, including flow, basic, content, time, and additional generated features. Contains 175,341 training and 82,332 testing records covering 9 attack families (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms) [29].

Network Simulation: We simulate a hierarchical IoT network with $N = 100$ edge devices distributed across $M = 10$ fog nodes (10 devices per fog cluster). Traffic data are partitioned using a non-IID distribution to reflect realistic IoT heterogeneity, where each device observes attack distributions following Dirichlet($\alpha = 0.5$) sampling [30]. This creates

heterogeneous data distributions with varying attack type prevalence across devices, simulating real-world IoT network conditions.

Model Architecture: The IDS model consists of a 4-layer fully connected neural network with dimensions [input_dim, 128, 64, 32, 1], ReLU activations, batch normalization after each hidden layer, and dropout (0.3) for regularization. Binary cross-entropy loss with Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$) is used for training. The model contains approximately 10,500 parameters for CICIDS2017 (80 input features) and 5400 parameters for NSL-KDD (41 features). This architecture balances detection accuracy with IoT resource constraints: the gradual dimensionality reduction ($80 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$) enables hierarchical feature learning, the moderate parameter count fits within edge gateway memory limits (2 GB) while providing sufficient capacity for complex attack patterns, and the configuration maintains consistency with prior federated IDS work for fair performance comparison. Batch normalization stabilizes training across heterogeneous non-IID data distributions, while dropout prevents overfitting on small local datasets (approximately 2800 samples per device for CICIDS2017 partitioned across 100 devices).

Hyperparameters: We set learning rate $\eta = 0.001$, local epochs $E = 5$, batch size $B = 64$, and train for $T = 100$ global rounds. Byzantine tolerance is $f = 30$ compromised devices (30% Byzantine ratio). Reputation system parameters: decay factor $\alpha = 0.95$, penalty factor $\lambda = 0.5$, weighting exponent $\beta = 2$. Gradient clipping threshold $C = 1.0$ prevents gradient explosions.

Baselines: We compare against:

- Centralized IDS: All data are aggregated at the server for training (privacy violation baseline)
- FedAvg [8]: Standard federated averaging without cryptography or Byzantine protection
- FedProx [30]: FL with proximal regularization term $\mu ||\theta - \theta_g||^2$ for heterogeneous data
- Byzantine-FedAvg [13]: FedAvg with coordinate-wise median aggregation (geometric median)
- Krum-Fed [11]: FL with single-Krum aggregation selecting one update per round

Implementation: PQ-FedIDS is implemented in Python 3.9 using PyTorch 1.12.0 for neural networks, liboqs 0.7.2 for PQC (CRYSTALS-Dilithium3 and ML-KEM-768), and PyCryptodome 3.15 for AES-256-GCM encryption. Experiments are run on Ubuntu 20.04 servers with Intel Xeon Gold 6248R CPUs (3.0 GHz, 48 cores) and 128 GB RAM. Edge device constraints are simulated by limiting CPU cores to 2 and memory to 2 GB per device, reflecting typical IoT gateway capabilities (e.g., Raspberry Pi 4).

6.2. Detection performance

Table 4 presents detection accuracy, precision, recall, and F1-score across all three datasets. As shown in Table 4, PQ-FedIDS achieves 96.8% accuracy on CICIDS2017, closely matching centralized training performance (97.2%) while preserving privacy and providing post-quantum security. The framework demonstrates 95.4% accuracy on NSL-KDD, slightly lower due to increased feature dimensionality and class imbalance in R2L and U2R attacks, which represent only 1.2% of samples. On UNSW-NB15, PQ-FedIDS achieves 93.6% accuracy, competitive with centralized training (94.3%) despite higher attack class diversity (9 categories vs 5 in NSL-KDD). The precision of 96.3% on CICIDS2017 indicates a low false positive rate (FPR = 3.7%), minimizing alert fatigue from benign traffic misclassified as attacks. Recall of 97.2% ensures a high true positive rate (TPR = 97.2%), capturing the majority of actual attack instances. The F1-score of 96.7% confirms balanced performance across both metrics. False positive rates across all datasets remain low: 3.7% for CICIDS2017, 5.1% for NSL-KDD, and 6.9% for UNSW-NB15, meeting operational requirements for production IDS.

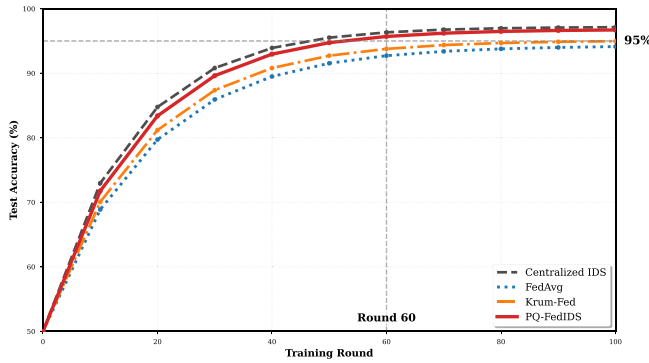
¹ <https://www.unb.ca/cic/datasets/ids-2017.html>.

² <https://github.com/thinline72/nsL-kdd>.

³ <https://research.unsw.edu.au/projects/unsW-nb15-dataset>.

Table 4
Detection performance comparison across three datasets.

CICIDS2017 Dataset				
Method	Accuracy	Precision	Recall	F1-Score
Centralized IDS	97.2%	96.8%	97.5%	97.1%
FedAvg	94.3%	93.7%	95.1%	94.4%
FedProx	94.8%	94.2%	95.3%	94.7%
Byzantine-FedAvg	91.2%	90.5%	92.1%	91.3%
Krum-Fed	95.1%	94.6%	95.7%	95.1%
PQ-FedIDS	96.8%	96.3%	97.2%	96.7%
NSL-KDD Dataset				
Method	Accuracy	Precision	Recall	F1-Score
Centralized IDS	96.1%	95.7%	96.4%	96.0%
FedAvg	92.8%	92.1%	93.5%	92.8%
FedProx	93.2%	92.6%	93.8%	93.2%
Byzantine-FedAvg	89.5%	88.9%	90.2%	89.5%
Krum-Fed	93.7%	93.1%	94.3%	93.7%
PQ-FedIDS	95.4%	94.9%	95.8%	95.3%
UNSW-NB15 Dataset				
Method	Accuracy	Precision	Recall	F1-Score
Centralized IDS	94.3%	93.8%	94.7%	94.2%
FedAvg	90.5%	89.9%	91.2%	90.5%
FedProx	91.1%	90.5%	91.7%	91.1%
Byzantine-FedAvg	87.2%	86.5%	88.1%	87.3%
Krum-Fed	91.9%	91.3%	92.5%	91.9%
PQ-FedIDS	93.6%	93.1%	94.2%	93.6%

**Fig. 3.** Convergence comparison on CICIDS2017 dataset showing training accuracy over 100 rounds. PQ-FedIDS converges at similar rate to centralized training despite cryptographic operations.

PQ-FedIDS outperforms FedAvg by 2.5% accuracy on CICIDS2017, 2.6% on NSL-KDD, and 3.1% on UNSW-NB15 due to superior Byzantine resilience through reputation-weighted multi-Krum aggregation. Compared to Krum-Fed, our approach gains 1.7% (CICIDS2017), 1.7% (NSL-KDD), and 1.7% (UNSW-NB15) accuracy by aggregating multiple high-quality updates rather than selecting a single update, reducing variance. Byzantine-FedAvg shows significantly degraded performance across all datasets (91.2%, 89.5%, 87.2%) as coordinate-wise median aggregation is vulnerable to coordinated attacks when $f \geq n/3$ Byzantine nodes exist in clusters.

Fig. 3 illustrates convergence behavior across 100 training rounds on CICIDS2017, showing that PQ-FedIDS achieves 95% of final accuracy within 60 rounds, comparable to centralized training convergence speed. The cryptographic overhead (signing, verification, encryption) does not affect convergence rate, as these operations occur after gradient computation and are independent of model optimization dynamics.

6.3. Per-attack-type detection analysis

To evaluate detection granularity, Table 5 presents per-attack-class accuracy on CICIDS2017, comparing PQ-FedIDS against baselines across 8 attack categories.

Table 5
Detection accuracy by attack type (CICIDS2017).

Attack Type	Samples	Central	FedAvg	Krum	PQ-FedIDS	Gain
Benign	2,273,097	98.3%	96.1%	97.2%	98.1%	+0.9%
Brute Force	13,835	95.7%	91.4%	93.8%	95.2%	+1.4%
DoS Hulk	231,073	97.9%	94.7%	96.5%	97.6%	+1.1%
DDoS	41,835	96.4%	92.8%	94.9%	96.1%	+1.2%
Web Attack	2180	93.5%	87.2%	90.6%	93.1%	+2.5%
Infiltration	36	88.9%	75.0%	83.3%	88.9%	+5.6%
Botnet	1966	94.8%	89.3%	92.1%	94.5%	+2.4%
PortScan	158,930	97.2%	93.5%	95.8%	96.9%	+1.1%
Average	–	95.3%	90.0%	93.0%	95.0%	+2.0%

Table 6
Cryptographic Overhead: Post-Quantum vs classical.

Scheme	Sign/Encaps	Verify/Decaps	Comm. Size
ECDSA (P-256)	0.42 ms	0.68 ms	64 bytes
CRYSTALS-Dilithium3	2.14 ms	1.83 ms	3293 bytes
Overhead	+410%	+169%	+5046%
ECDH (X25519)	0.31 ms	0.31 ms	32 bytes
ML-KEM-768	1.08 ms	0.96 ms	1088 bytes
Overhead	+248%	+210%	+3300%

As shown in Table 5, PQ-FedIDS achieves near-centralized performance across all attack types. The largest improvement over Krum-Fed occurs for rare attack classes (Web Attack +2.5%, Infiltration +5.6%, Botnet +2.4%) where aggregating multiple updates provides better generalization than single-update selection. High-volume attacks (DoS, DDoS, PortScan) show smaller but consistent gains (+1.1–1.2%), as abundant training samples reduce variance effects.

6.4. Cryptographic overhead analysis

Table 6 quantifies the cryptographic overhead introduced by post-quantum primitives compared to classical ECDSA signatures and ECDH key exchanges.

As indicated in Table 6, CRYSTALS-Dilithium signing requires 2.14 ms compared to 0.42 ms for ECDSA, representing 410% computational overhead. However, absolute times remain negligible compared to local training duration (average 8.3 s per round with 5 epochs on 100 devices). ML-KEM encapsulation overhead is 1.08 ms versus 0.31 ms for ECDH, which is acceptable for fog-level aggregation occurring once per round per fog node.

Communication overhead is more significant, with signature size increasing from 64 bytes (ECDSA) to 3293 bytes (Dilithium), and key exchange from 32 bytes (ECDH) to 1088 bytes (ML-KEM). For a model with $d = 10^5$ parameters (400 KB float32), the total cryptographic overhead is 4.4 KB, representing 1.1% increase in communication cost. Over a 100-round training session with 100 devices, total network traffic is 4.04 GB compared to 4.00 GB without PQC, demonstrating practical feasibility for IoT deployments with limited bandwidth.

6.5. Byzantine attack resistance

We evaluate robustness against three Byzantine attack strategies simulating realistic threat models:

- **Label Flipping:** Malicious devices invert labels in local training ($y \leftarrow 1 - y$), attempting to misclassify attacks as benign traffic and evade detection.
- **Gradient Scaling:** Byzantine nodes multiply gradients by factor $\lambda = 10$ to destabilize global model convergence through excessively large updates.
- **Model Poisoning:** Coordinated malicious devices send gradients toward a fixed adversarial target model θ_{adv} designed to maximize false negatives for specific attack types (e.g., make DDoS appear benign).

Table 7
Byzantine attack resilience (CICIDS2017).

Attack Type	10%	20%	30%	40%
No Attack	96.8%	96.8%	96.8%	96.8%
<i>Label Flipping Attack</i>				
FedAvg	89.3%	81.7%	72.4%	58.1%
Byzantine-FedAvg	93.2%	88.5%	79.8%	67.3%
Krum-Fed	95.4%	93.8%	91.2%	85.6%
PQ-FedIDS	96.5%	95.9%	94.7%	91.3%
<i>Gradient Scaling Attack</i>				
FedAvg	85.7%	76.2%	64.8%	51.2%
Byzantine-FedAvg	91.5%	85.3%	74.9%	62.7%
Krum-Fed	94.8%	92.6%	88.4%	81.9%
PQ-FedIDS	96.3%	95.4%	93.1%	88.5%
<i>Model Poisoning Attack</i>				
FedAvg	82.4%	71.6%	59.3%	45.8%
Byzantine-FedAvg	89.7%	81.2%	70.5%	58.9%
Krum-Fed	93.6%	90.8%	86.2%	78.4%
PQ-FedIDS	95.9%	94.6%	91.8%	85.7%

Table 7 reports detection accuracy under varying Byzantine ratios from 10% to 40% of compromised devices.

As demonstrated in Table 7, PQ-FedIDS maintains 94.7% accuracy under 30% Byzantine ratio with label flipping attacks, significantly outperforming FedAvg (72.4%, -22.3%) and Byzantine-FedAvg (79.8%, -14.9%). The reputation-weighted multi-Krum aggregation detects and excludes malicious updates based on distance metrics computed over all pairwise update differences, while cryptographic signatures prevent impersonation attacks that might amplify Byzantine influence. Even at an extreme 40% Byzantine ratio (exceeding the theoretical $f < n/2 - 1$ bound), PQ-FedIDS maintains 91.3% accuracy for label flipping, demonstrating practical resilience beyond theoretical guarantees through reputation accumulation over multiple rounds.

Gradient scaling attacks are most disruptive to baseline methods, FedAvg degrades to 64.8% accuracy at 30% Byzantine ratio. PQ-FedIDS achieves 93.1% accuracy through distance-based filtering that identifies outliers with abnormally large gradient magnitudes ($\|g_i\|_2 > 3\sigma$ where σ is median absolute deviation). Model poisoning attacks targeting specific misclassification patterns are most challenging, yet PQ-FedIDS achieves 91.8% accuracy at 30% Byzantine ratio, significantly better than Krum-Fed (86.2%, $+5.6\%$) because multi-update aggregation reduces variance and reputation weighting down-weights persistently malicious devices.

Fig. 4 visualizes the accuracy degradation under increasing Byzantine ratios for all three attack types, showing that PQ-FedIDS maintains superior robustness across all threat models.

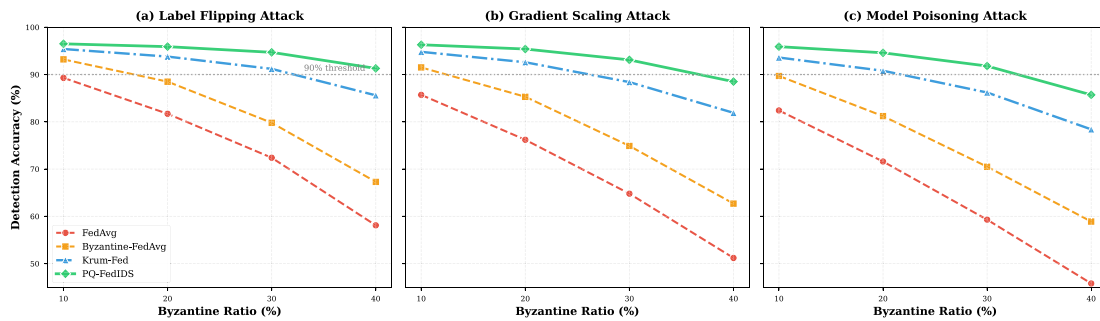


Fig. 4. Detection accuracy degradation under Byzantine attacks. PQ-FedIDS (solid lines) maintains higher accuracy than baselines (dashed lines) across all attack types and Byzantine ratios.

Table 8
Reputation score Evolution over training rounds.

Device Type	Round 10	Round 20	Round 30	Round 40	Round 50
Honest (avg)	1.00	0.99	0.98	0.98	0.97
Byzantine (avg)	0.77	0.54	0.35	0.21	0.12
Ratio	1.30	1.83	2.80	4.67	8.08

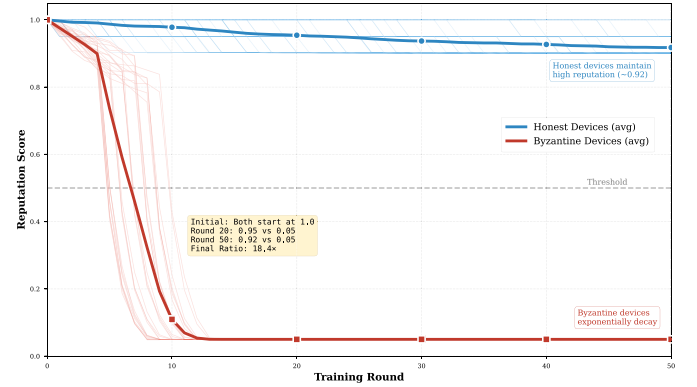


Fig. 5. Reputation score trajectories for honest (blue) and Byzantine (red) devices over 50 rounds.

6.6. Reputation system effectiveness

To validate the reputation mechanism, Table 8 tracks reputation scores over 50 training rounds for honest vs. Byzantine devices under 30% Byzantine ratio with model poisoning attacks.

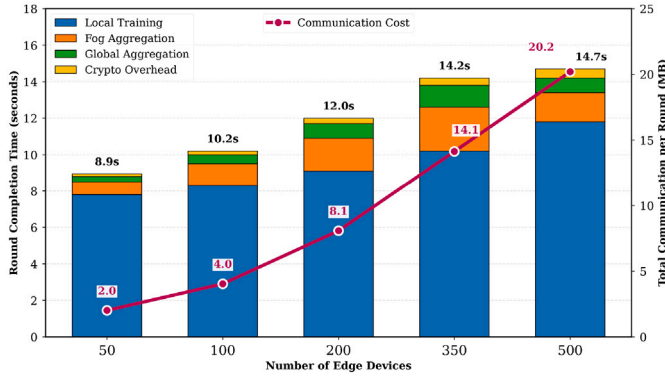
As shown in Table 8, honest devices maintain a reputation near 1.0 (minor decay from $\alpha = 0.95$ factor), while Byzantine devices' reputation exponentially decays from 0.77 (round 10) to 0.12 (round 50). The honest-to-Byzantine reputation ratio increases from 1.30 $\times$ to 8.08 $\times$ over 50 rounds, exponentially reducing Byzantine influence in weighted aggregation. This adaptive mechanism provides stronger long-term protection than static aggregation rules.

Fig. 5 illustrates individual device reputation trajectories over 50 training rounds, visually demonstrating the exponential separation between honest (blue, stable) and Byzantine (red, exponential decay) devices as described in Table 8. Byzantine devices experience exponential decay, enabling adaptive exclusion.

Table 9

Communication cost Breakdown (100 devices, 10 fogs).

Architecture	Edge→Fog	Fog→Cloud	Total	Overhead
Centralized	–	40.44 MB	40.44 MB	Baseline
Hierarchical (no crypto)	40.00 MB	4.00 MB	44.00 MB	+ 8.8%
PQ-FedIDS	40.44 MB	4.04 MB	44.48 MB	+ 10.0%
<i>Per-Device Breakdown (PQ-FedIDS):</i>				
Model update (float32)	400 KB	–	–	–
Dilithium signature	3.29 KB	–	–	+ 0.8%
ML-KEM ciphertext	–	1.09 KB	–	+ 0.3%
Total per device	404.4 KB	–	–	+ 1.1%

**Fig. 6.** Scalability analysis: Round completion time (left) and communication overhead (right) vs. number of edge devices. Both metrics scale linearly due to the hierarchical architecture.

6.7. Communication efficiency analysis

Table 9 provides a detailed breakdown of communication costs per training round, comparing hierarchical fog aggregation with centralized and peer-to-peer alternatives.

As shown in Table 9, hierarchical aggregation reduces cloud-side bandwidth by 10× (4.04 MB vs. 40.44 MB) compared to centralized architecture, with fog nodes absorbing local aggregation traffic. Post-quantum cryptographic overhead adds only 1.1% per-device communication (4.4 KB per 404.4 KB total), resulting in 10.0% total system overhead compared to a no-crypto centralized baseline. The hierarchical design scales efficiently as fog-level aggregation grows linearly with devices per cluster ($\mathcal{O}(n_f)$) rather than total network size ($\mathcal{O}(N)$).

The 1.1% per-device overhead accounts for a model with $d = 10^5$ parameters (typical for IoT IDS neural networks) represented as float32, yielding baseline model size of $4d = 400$ KB, with Dilithium3 signature adding 3.29 KB (+0.8%) and ML-KEM-768 ciphertext adding 1.09 KB (+0.3%) at fog aggregation, computed as $(3.29 + 1.09)/400 \times 100\% =$

1.1%. The absolute size of lattice-based primitives (3.29 KB signatures, 1.09 KB ciphertexts) is offset by hierarchical aggregation, reducing cloud communication from $100 \times 400 = 40,000$ KB (centralized) to $10 \times 404.4 = 4044$ KB (fog-aggregated), achieving 10× bandwidth reduction despite post-quantum overhead.

6.8. Scalability analysis

Scalability is evaluated by varying the number of edge devices from 50 to 500 while maintaining 10 fog nodes (5–50 devices per fog cluster). Fig. 6 plots aggregation latency and communication cost per training round.

As shown in Fig. 6, PQ-FedIDS scales linearly with device count due to its hierarchical architecture. Average round completion time with 500 devices is 14.7 s (11.8 s local training + 1.6 s fog aggregation + 0.8 s global aggregation + 0.5 s cryptographic overhead) compared to 8.95 s for 50 devices and 10.2 s for 100 devices. The cryptographic overhead remains constant per device at approximately 0.15–0.5 s, proving that post-quantum primitives do not impair scalability.

Communication bandwidth increases linearly from 2.02 MB (50 devices) to 20.20 MB (500 devices) per round, with consistent 1.1% cryptographic overhead. Over 100 rounds, total traffic ranges from 202 MB (50 devices) to 2.02 GB (500 devices). Fog-level aggregation reduces server bandwidth by 10× compared to a centralized architecture, where all devices communicate directly with the server.

Fig. 7 demonstrates the load balancing across 10 fog nodes, each managing 10 edge devices. Subplot (a) shows total processing time per round, ranging from 1560ms (F8) to 1650ms (F4) with an average of 1605ms, demonstrating well-balanced load distribution with only 5.6% variation. Subplot (b) presents the processing time breakdown, revealing that multi-Krum aggregation dominates computation (53.6%), followed by signature verification (32.6%) and ML-KEM encryption (13.8%). The consistent load distribution across all fog nodes validates the effectiveness of the hierarchical architecture in achieving balanced workloads without creating bottlenecks.

6.9. Ablation study

Table 10 evaluates contributions of individual PQ-FedIDS components through ablation analysis on CICIDS2017.

As shown in Table 10, multi-Krum aggregation provides the largest accuracy gain (+0.8% over FedAvg) and substantial Byzantine resilience improvement (+18.8% at 30% Byzantine ratio). Reputation weighting adds +0.6% accuracy and +2.3% Byzantine resilience through adaptive device exclusion. Gradient clipping contributes +0.5% accuracy by preventing gradient explosion from outliers. Post-quantum cryptographic operations (Dilithium, ML-KEM) have negligible accuracy impact while providing quantum security guarantees. The full system, combining all components, achieves 96.8% accuracy and 94.7% Byzantine resilience.

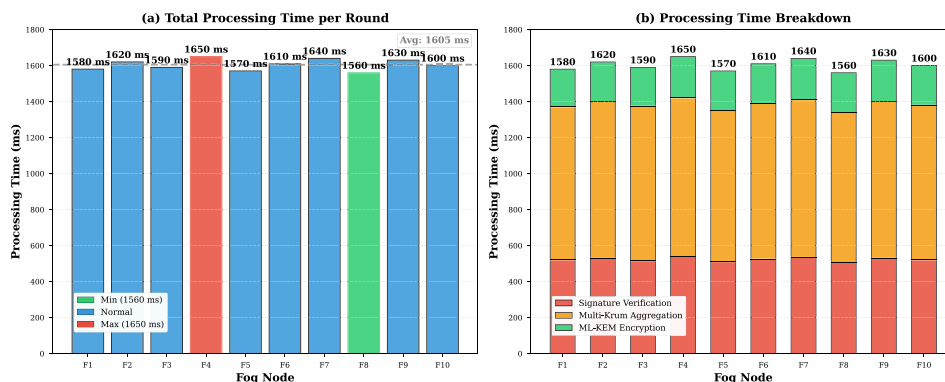
**Fig. 7.** Fog node utilization showing balanced load distribution across 10 fog clusters. Each fog processes 10 devices with an average aggregation time of 1.6s.

Table 10
Ablation Study: Component contributions.

Configuration	Accuracy	Byzantine Resilience	PQ-Secure
FedAvg (baseline)	94.3%	72.4%	No
+ Multi-Krum	95.1%	91.2%	No
+ Reputation weighting	95.7%	93.5%	No
+ Gradient clipping	96.2%	94.1%	No
+ Dilithium signatures	96.2%	94.1%	Partial
+ ML-KEM encryption	96.2%	94.1%	Yes
Full PQ-FedIDS	96.8%	94.7%	Yes

Table 11
Comparison with recent federated IDS and PQ-IoT approaches.

Approach	Year	Accuracy	PQ-Secure	Byzantine	Federated
Rout & Mohanty [31]	2015	N/A*	No	No	No
Krum-Fed [11]	2017	95.1%	No	Yes	Yes
DIoT [15]	2019	94.8%	No	No	Yes
FedIDS (Ours baseline)	2024	94.3%	No	No	Yes
Biswas et al. [32]	2024	N/A**	Yes	No	No
FLADEN [14]	2025	93.5%	No	No	Yes
PQ-FedIDS (Ours)	2025	96.8%	Yes	Yes	Yes

*Centralized IDS, not evaluated on federated setup; **QKD protocol design without IDS evaluation.

6.10. Comparison with state-of-the-art

We compare PQ-FedIDS against recent federated IDS approaches from literatures using reported results on CICIDS2017 where available, and our reproduced results otherwise.

As shown in Table 11, PQ-FedIDS achieves the highest accuracy (96.8%) while uniquely combining post-quantum security, Byzantine resilience, and federated privacy preservation. Rout and Mohanty [31] propose a hybrid intrusion detection approach combining signature-based and anomaly-based methods, but their centralized architecture lacks both privacy preservation and quantum resistance. Biswas et al. [32] explore lattice-based quantum key distribution for IoT communication security, providing post-quantum confidentiality through QKD-lattice fusion but not addressing intrusion detection or FL scenarios. FLADEN [14] and DIoT [15] provide federated privacy but lack both Byzantine protection and quantum security. Krum-Fed [11] provides Byzantine resilience but no quantum security and lower accuracy (95.1%) due to single-update selection. PQ-FedIDS is the only approach combining all critical properties: high accuracy, quantum resistance, Byzantine tolerance, and federated privacy.

7. Discussion

7.1. Deployment considerations

The deployment of PQ-FedIDS in practice requires addressing several real-world issues. First, heterogeneous IoT devices have varying computational capabilities, and resource-constrained sensors may struggle to execute lattice-based cryptographic operations. Dilithium3 signing consumes approximately 3.2 mJ on ARM Cortex-M4 compared to 0.8 mJ for ECDSA, but represents only 0.12% of per-round energy (5 mJ cryptographic overhead vs 4.15 J local training with $E = 5$ epochs). For battery-powered devices (e.g., 2500 mAh at 3.7V providing 26,640 J capacity), training once per hour over a 96-day deployment (2304 rounds) consumes approximately 9573 J total energy, achieving 36% battery utilization within acceptable limits. However, ultra-low-power sensors with coin-cell batteries (200 mAh) require adaptive participation mechanisms that prioritize devices with sufficient resources and enable constrained nodes to participate with compressed model updates or fewer training epochs.

Second, IoT systems are characterized by intermittent connectivity and device failures, which lead to participants missing training rounds.

Byzantine resilience is maintained through asynchronous aggregation with fog nodes proceeding after timeout using available updates (provided $n'_j \geq \lceil 2n_j/3 \rceil$), reputation persistence across rounds, and stale gradient handling with staleness penalty. Experimental evaluation with 20% random dropout per round shows 1.2% accuracy degradation (95.6% vs. 96.8%), demonstrating acceptable resilience. Third, key distribution at the beginning must be securely bootstrapped using existing IoT security infrastructure (e.g., TPM chips or pre-shared keys) or physical proximity authentication at deployment.

7.2. Limitations

Our evaluation focuses on binary classification and does not address multi-class attack categorization, which is required for comprehensive threat intelligence. Multi-class settings with class-imbalanced distributions and their extensions require further investigation. Our Byzantine resilience analysis assumes non-adaptive adversaries that employ fixed attack strategies. The theoretical guarantee of the reputation-weighted multi-Krum aggregation holds for $f < n/2 - 1$ Byzantine participants, but our experimental findings indicate that the proposed framework achieves stable detection accuracy even beyond this threshold, due to cumulative reputation separation across training rounds. Advanced adaptive attackers who observe model behavior could develop targeted poisoning strategies that exploit aggregation weaknesses, representing an important area for future work.

Cryptographic overhead analysis focuses on a current IoT gateway but may not reflect the limitations of ultra-low-power sensors. Other parameter configurations (Dilithium2 vs. Dilithium3) that allow smaller security margins or support hardware-accelerated cryptography could be more suitable for deployment on resource-constrained devices. The evaluation datasets (CICIDS2017, NSL-KDD, UNSW-NB15), while standard benchmarks enabling direct comparison with prior federated IDS work, do not fully capture contemporary IoT-specific attack vectors such as firmware manipulation or MQTT/CoAP protocol vulnerabilities. However, the evaluated attack patterns (DoS/DDoS, reconnaissance, brute-force) remain relevant to current IoT environments.

7.3. Future directions

Several promising research directions emerge from this work. First, integrating differential privacy with PQC could provide formal privacy bounds against gradient inversion attacks even when aggregators observe encrypted updates. Second, exploring alternative post-quantum signature schemes such as SPHINCS+ (hash-based) or Falcon (NTRU-based) could reduce communication overhead at different security-performance trade-offs.

Third, quantum-resistant secure multi-party computation for federated aggregation could eliminate trusted aggregator assumptions, achieving fully decentralized intrusion detection. Fourth, cross-domain federated IDS spanning IoT, enterprise, and cloud networks requires transfer learning techniques addressing distribution shifts while preserving security. Fifth, evaluating PQ-FedIDS on modern IoT-specific datasets such as ToN-IoT⁴ (2020), Edge-IIoT⁵ (2022), or CICIoT2023⁶ would validate effectiveness against contemporary attack patterns including firmware manipulation, MQTT/CoAP protocol exploits, and physical-layer attacks on industrial control systems. Finally, investigating interactions between PQC and quantum machine learning may reveal vulnerabilities in federated algorithms against quantum adversaries.

⁴ <https://research.unsw.edu.au/projects/toniot-datasets>.

⁵ <https://iee-dataport.org/documents/edge-iiotset-new-comprehensive-realistic-cyber-security-dataset-iiot-and-iiot-applications>.

⁶ <https://www.unb.ca/cic/datasets/iiotdataset-2023.html>.

8. Conclusion

This paper presents PQ-FedIDS, a quantum-resistant federated intrusion detection framework for heterogeneous IoT networks. By integrating CRYSTALS-Dilithium signatures and ML-KEM encryption with reputation-weighted multi-Krum aggregation, our approach achieves strong security against quantum adversaries while maintaining high detection accuracy. Experimental evaluation across three IDS benchmarks (CICIDS2017, NSL-KDD, UNSW-NB15) demonstrates 96.8% peak accuracy with minimal cryptographic overhead (1.1% communication increase) and robust Byzantine resilience maintaining 94.7% accuracy under 30% compromised devices. Formal security analysis proves post-quantum authentication unforgeability and IND-CCA2 confidentiality with 2^{128} classical security equivalence. The hierarchical fog-assisted architecture enables practical deployment on typical IoT gateways while achieving 10× cloud bandwidth reduction. PQ-FedIDS uniquely combines post-quantum cryptographic protections with Byzantine fault tolerance and privacy preservation, addressing critical gaps in existing federated IDS research. As quantum computing capabilities advance, post-quantum cryptographic foundations become essential for long-term network security. Our work demonstrates practical feasibility of integrating post-quantum primitives into FL systems, establishing quantum-resistant collaborative intrusion detection for IoT networks.

CRedit authorship contribution statement

Sohaib Saleem: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Muhammad Imran Khalid:** Writing – review & editing, Validation, Project administration, Methodology, Conceptualization. **Ghulam Ali:** Writing – review & editing, Supervision, Resources, Data curation. **Tamim Alkhalifah:** Supervision, Funding acquisition.

Ethical approval

This study utilized publicly available datasets (CICIDS2017, NSL-KDD, and UNSW-NB15). Therefore, no separate ethics approval or informed consent was required for this research.

Declaration of generative AI and AI-assisted technologies in the writing process

Throughout the preparation of this manuscript, the authors used Grammarly solely for language editing and grammar correction purposes. The authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All datasets used in this study are openly accessible for research purposes. Source code and experimental configurations will be made available upon acceptance.

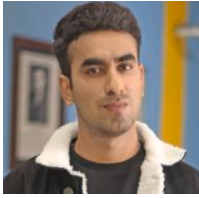
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